THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 9: **Methods of Separation in Everyday Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A mixture contains chalk powder, salt and iron filings. The first method to use is: 1

(a) filtration **(b)** evaporation **(c)** magnet **(d)** sieving

2. A solution is saturated at 30 °C. On heating to 50 °C it becomes: 1

(a) more saturated **(b)** gas **(c)** solid **(d)** unsaturated

3. Why do we heat the salt solution slowly towards the end during evaporation? 1

(a) to freeze **(b)** to cool

(c) to dissolve more **(d)** to avoid spurting

4. After removing iron and adding water to the mixture of chalk and salt, filtering gives chalk as: 1

(a) residue **(b)** solvent **(c)** filtrate **(d)** solute

5. When a hot saturated solution cools, the excess solute: 1

(a) dissolves more **(b)** crystallises out **(c)** becomes gas **(d)** evaporates

6. Why is the filter paper folded and fitted in the funnel? 1

(a) to cool **(b)** to make it colourful

(c) to hold the paper well **(d)** to prevent filtration

7. After the filtration above, the filtrate contains: 1

(a) sand **(b)** salt solution **(c)** iron **(d)** chalk

8. A student added 5 spoons of salt to 100 mL of water and 3 spoons dissolved. This means: 1

(a) solution became saturated **(b)** no salt dissolved

(c) water evaporated **(d)** all dissolved

9. What happens to the water in a closed container heated and its vapour cooled? 1

(a) air is collected **(b)** sand is collected

(c) pure water is collected **(d)** salt is collected

10. To get salt from the filtrate we use: 1

(a) evaporation **(b)** sieving **(c)** winnowing **(d)** magnet

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is evaporation unsuitable to get sand from a sand-water mixture? 2

12. Why can we not separate a mixture of salt and sugar by filtration? 2

13. Why is the mud left on the filter paper called residue? 2

14. How do you separate a mixture of sugar and iron filings? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A mixture contains wheat, stones, iron nails and sand. Suggest a plan to separate. 3

16. Explain why sea water is evaporated in shallow pans instead of deep ones. 3

17. Why can a sieve not separate chalk powder from water? 3

OR

Describe the experiment to prove that a salt solution has salt even if it looks clear. 3

18. What is the difference between a pure substance and a mixture? 3

OR

How do you decide whether to use filtration or evaporation to separate a solid from water? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Plan the separation of a mixture containing sand, sugar, iron filings and water-insoluble sawdust. 5

OR

Analyse how the separation methods used in cooking and farming are related. 5

20. Describe an experiment to prove that the amount of solute dissolved depends on temperature. 5

OR

Describe how to recover a dissolved solid and the water from a solution at the same time. 5

END OF QUESTION PAPER — Total: 40 Marks

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